

Preprint

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Abstract 論文のあらまし：1.（新規）今回の線形反応に高次の反応（保存量を破壊しない）があっても、収束速度（リアプノフ指数）は最大固有値で評価可能。2. 線形の場合で示した E（無限大）の最大固有（リアプノフ指数）の振る舞いをネットワーク条件で書き直し。3.（新規）2の結果から直ぐに加減速性を特徴づけることのできる条件を5つほど掲載（前回の環状やコバルトの結果を拡張したものもあります。大岡さんから見て意味のありそうなもののみを掲載したらよいかなと思いました。系4. 4, 4. 5は結構面白いかもしれません）。4. 適用例を載せたらよいと思います。少なくとも、具体的な反応系に対してリアプノフ指数と固有値が一致することと加減速性は述べたほうが良いかと思いました。＊今のところ数学の結果のみを記述しているので、投稿先にもよりますが随所に物理的な説明の追記が必要。

Keywords.

AMS subject classifications.

1 Introduction

2 Nonlinear Chemical Reaction Network Model of an Electrocatalyst undergoing Deactivation

We consider a chemical reaction network used to model an electrocatalyst undergoing deactivation. Let the chemical reaction network be mathematically defined by a triplet $(S, C, \mathcal{R})$, where the elements S , C , and $\mathcal{R}$ are defined by [20]:

S : the set of n intermediate species involved in the catalytic cycle, which are denoted by $X_1, X_2, \dots, X_n$,

C : the set of all *complexes* y in the network,

$\mathcal{R}$: the set of all *chemical reactions* $y \rightarrow y'$ in the network.

Here a complex means a linear combination of species, $y = y_1X_1 + \dots + y_nX_n$, with coefficients of non-negative integers, $y_1, y_2, \dots, y_n$, and a chemical reaction $y \rightarrow y'$ denotes that a complex y' is produced from another complex y . In particular, we may associate y with the vector of these coefficients,

34 $(y_1, y_2, \dots, y_n)^T$, and y and y' are called a *reactant* and a *product*, respectively.

35 We partition the reaction set $\mathcal{R}$ into two subsets $\mathcal{R}_1$ and $\mathcal{R}_{nl}$:

$$\mathcal{R} = \mathcal{R}_1 \cup \mathcal{R}_{nl} \quad (1)$$

36 The subset $\mathcal{R}_1$ comprises all first-order catalytic and dissolution steps. A first-order catalytic step from
 37 species X_i to X_j is represented as $X_i \rightarrow X_j$. A dissolution step, representing the irreversible loss of
 38 species X_i into the electrolyte, is denoted by $X_i \rightarrow \emptyset$, where $\emptyset$ signifies a state external to the catalytic
 39 cycle. The topology of the first-order sub-network $(\mathcal{S}, \mathcal{R}_1)$, excluding dissolution pathways, is described
 40 by a weighted directed graph $G := (\mathcal{V}, \mathcal{E})$, where the vertex set $\mathcal{V} = \{X_1, \dots, X_n\}$ corresponds to the
 41 intermediate species and the edge set $\mathcal{E} \subset \mathcal{R}_1$ represents the first-order catalytic reactions [11, 36].
 42 Throughout the present paper, we assume that the graph G is strongly connected, that is, for any reaction
 43 $X_i \rightarrow X_j \in \mathcal{E}$ there exist s species X_{i_p} ($i_p \in \{1, \dots, n\}$, $p = 1, \dots, s$) such that all of the reactions
 44 $X_j \rightarrow X_{i_1}$, $X_{i_1} \rightarrow X_{i_2}$, $\dots$, $X_{i_s} \rightarrow X_i$ are contained in $\mathcal{E}$. We also assume the presence of at least one
 45 dissolution reaction ($X_i \rightarrow \emptyset$) within the network [11, 36].

46 On the other hand, the subset $\mathcal{R}_{nl}$ incorporates the higher-order nonlinear reactions. For every reaction
 47 $y \rightarrow y' \in \mathcal{R}_{nl}$, the reactant complex satisfies the following condition:

$$\sum_{i=1}^n y_i \geq 2 \quad (2)$$

48 The condition (2) represents reactions involving multiple intermediate species as reactants.

49 Now, let us consider how the amounts of catalytic intermediates change with time. Let us use $x_i \in \mathbb{R}_{\geq 0}$
 50 to denote the amount of species X_i , and define a state vector $x := (x_1, x_2, \dots, x_n)^T \in \mathbb{R}_{\geq 0}^n$, where $\mathbb{R}_{\geq 0}$ is
 51 the non-negative orthant of $\mathbb{R}$. For a typical electrocatalyst, intermediate species are confined to the two
 52 dimensional surface of the solid and therefore each X_i has units of $[\text{mol}/\text{cm}^2]$. However, units do not in-
 53 fluence the mathematical analysis below, and our analysis remains applicable even if some intermediates
 54 freely diffuse in 3 dimensional space, such as in the case of homogeneous or enzyme catalysis. Then,
 55 the time-evolution of concentrations x can be described by the following ordinary differential equation
 56 (ODE) decomposed into a linear term and a nonlinear term [20]:

$$\frac{d}{dt}x(t) = A(E)x(t) + F(x(t)). \quad (3)$$

57 The linear term of the ODE (3) corresponding to the reactions $\mathcal{R}_1$ is given by the coefficient matrix
 58 $A(E) := L - D \in \mathbb{R}^{n \times n}$. Here, $L := [l_{ij}(E)]_{1 \leq i, j \leq n} \in \mathbb{R}^{n \times n}$ is the Laplacian matrix of the graph G defined
 59 by [21, 22]:

$$l_{ij}(E) = \begin{cases} k_{ji}(E) & (j \neq i) \\ -\sum_{p=1}^n k_{ip}(E) & (j = i) \end{cases} \quad (i, j = 1, \dots, n), \quad (4)$$

60 and $D \in \mathbb{R}^{n \times n}$ is the non-zero diagonal matrix:

$$D := \text{diag}(k_{10}(E), \dots, k_{n0}(E)) \neq O_n, \quad (5)$$

61 where $O_n \in \mathbb{R}^{n \times n}$ is the matrix of which all components are zero. The parameters $k_{ij}(E)$ represent the rate
 62 constants for the first-order reactions $X_i \rightarrow X_j$ and $X_i \rightarrow \emptyset$. Under potentiostatic control, the relationship
 63 between the electrode potential E and each rate constant follows the Butler-Volmer equation [24]:

$$k_{ij}(E) := k_{ij}^0 \exp(\alpha_{ij}E), \quad i \neq j, \quad i = 1, \dots, n, \quad j = 0, \dots, n \quad (6)$$

64 where $k_{ij}^0 \geq 0$ is the pre-exponential factor and $\alpha_{ij} \geq 0$ is the electron transfer coefficient. By these
 65 definitions, all off-diagonal elements of $A(E)$ are non-negative ($a_{ij}(E) = k_{ji}(E) \geq 0$ for $i \neq j$), and its

column sums satisfy $\sum_{i=1}^n a_{ij}(E) = -k_{j0}(E) \leq 0$. Because the graph G is strongly connected and at least one irreversible dissolution pathway is present, the matrix $-A(E)$ is an irreducible M-matrix [23, 28]. Let $\lambda_1(E), \dots, \lambda_n(E) \in \mathbb{C}$ be the eigenvalues of $A(E)$. According to the spectral properties of such matrices [28, 21, 22], it holds that

$$0 > \operatorname{Re}(\lambda_1(E)) \geq \operatorname{Re}(\lambda_2(E)) \geq \dots \geq \operatorname{Re}(\lambda_n(E)). \quad (7)$$

In particular, $\lambda_1(E)$ is a real and simple root. Furthermore, there exists a strictly positive left eigenvector $\tilde{w} \in \mathbb{R}_{>0}^n$ satisfying

$$\tilde{w}^T A(E) = \lambda_1(E) \tilde{w}^T \quad (8)$$

The nonlinear term $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of the ODE (3) corresponding to the reactions $\mathcal{R}_{nl}$ is formulated as

$$F(x) = \sum_{y \rightarrow y' \in \mathcal{R}_{nl}} k_{y \rightarrow y'} x^y (y' - y), \quad (9)$$

where $k_{y \rightarrow y'} > 0$ is the reaction rate constant for each reaction $y \rightarrow y' \in \mathcal{R}_{nl}$ and x^y is defined by $x^y := x_1^{y_1} x_2^{y_2} \dots x_n^{y_n}$.

From Eq. (2), every term in (9) consists of polynomials of degree two or higher. This means that $\|F(x)\| = O(\|x\|^2)$ in the neighborhood of the origin, satisfying $F(0) = 0$ and $DF(0) = O$, where $DF(0)$ is the Jacobian matrix of F at the origin. Here, $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^n$, and the Landau symbol $O(\|x\|^2)$ indicates that there exists a constant $M > 0$ such that $\|F(x)\| \leq M\|x\|^2$ for all x sufficiently close to the origin. Moreover, we assume that for any reaction $y \rightarrow y' \in \mathcal{R}_{nl}$, the following condition holds with respect to the left eigenvector $\tilde{w} \in \mathbb{R}_{>0}^n$ of $A(E)$:

$$\tilde{w}^T (y' - y) \leq 0. \quad (10)$$

It has been proved that the ODE (3) is non-negative (resp. positive). That is, any solution $x(t)$ to (3) with an initial value $x(0) \in \mathbb{R}_{\geq 0}^n$ (resp. $x(0) \in \mathbb{R}_{>0}^n$) remains in $\mathbb{R}_{\geq 0}^n$ (resp. $\mathbb{R}_{>0}^n$) for all $t \geq 0$, where $\mathbb{R}_{>0}^n$ is the positive orthant of $\mathbb{R}^n$ [37].

3 Characterization of the Lyapunov Characteristic Exponent

In this section, we prove that any solution to the ODE (3) globally converges to the origin corresponding to the deactivation state, and the Lyapunov characteristic exponent, which determines the asymptotic deactivation rate of the solution to Eq. (3), is determined by the dominant eigenvalue $\lambda_1(E)$ of the linear term $A(E)$ of Eq. (3).

Theorem 3.1 For any initial value $x(0) \in \mathbb{R}_{\geq 0}^n$, the solution $x(t)$ globally converges to the origin corresponding to the deactivation state. That is, $\lim_{t \rightarrow \infty} \|x(t)\| = 0$.

Proof As established in Section 2, there exists a strictly positive left eigenvector $\tilde{w} \in \mathbb{R}_{>0}^n$ corresponding to $\lambda_1(E)$ satisfying Eq. (8). We define a Lyapunov function $V(x) := \tilde{w}^T x$. Since $\tilde{w}_i > 0$ for all $i = 1, \dots, n$, $V(x) \geq 0$ for all $x \in \mathbb{R}_{\geq 0}^n$, and $V(x) = 0$ if and only if $x = 0$. Therefore, from the equivalence of norms, there exist positive constants c_1, c_2 such that:

$$c_1 \|x\| \leq V(x) \leq c_2 \|x\|, \quad \forall x \in \mathbb{R}_{\geq 0}^n \quad (11)$$

Evaluating the time derivative of $V(x)$ along the solution $x(t)$, we can obtain

$$\frac{d}{dt} V(x(t)) = \tilde{w}^T \frac{d}{dt} x(t) = \tilde{w}^T (A(E)x(t) + F(x(t)))$$

$$\begin{aligned}
&= \tilde{w}^T A(E)x(t) + \tilde{w}^T \sum_{y \rightarrow y' \in \mathcal{R}_{nl}} k_{y \rightarrow y'} x^y(t)(y' - y) \\
&= \lambda_1(E) \tilde{w}^T x(t) + \sum_{y \rightarrow y' \in \mathcal{R}_{nl}} k_{y \rightarrow y'} x^y(t) \tilde{w}^T (y' - y), \quad \forall t \in [0, +\infty).
\end{aligned} \tag{12}$$

From Eq. (8), the condition (10) and $k_{y \rightarrow y'} x^y \geq 0$ for $x \in \mathbb{R}_{\geq 0}^n$, we obtain the following differential inequality:

$$\frac{d}{dt} V(x(t)) \leq \lambda_1(E) V(x(t)), \quad \forall t \in [0, +\infty). \tag{13}$$

Since $\lambda_1(E) < 0$ from (7), it follows from Grönwall's inequality that

$$V(x(t)) \leq V(x(0)) e^{\lambda_1(E)t}, \quad \forall t \in [0, +\infty). \tag{14}$$

Taking the limit as $t \rightarrow \infty$ in Eq. (14), we have $V(x(t)) \rightarrow 0$. Combined with Eq. (11), it holds that $\lim_{t \rightarrow \infty} \|x(t)\| = 0$. $\square$

To quantify the effective deactivation rate, we define the Lyapunov characteristic exponent $\Lambda(E; x(0))$ for a solution $x(t)$ with initial value $x(0)$ by [29]:

$$\Lambda(E; x(0)) := \lim_{t \rightarrow \infty} \frac{1}{t} \ln \|x(t)\| \tag{15}$$

When this limit exists, it ensures that for any arbitrarily small $\eta > 0$, there exists a sufficiently large time $T > 0$ such that the norm is exponentially bounded:

$$C_1 e^{(\Lambda(E; x(0)) - \eta)t} \leq \|x(t)\| \leq C_2 e^{(\Lambda(E; x(0)) + \eta)t}, \quad \forall t > T \tag{16}$$

where C_1 and C_2 are positive constants.

Based on this definition (15), we show the following theorem regarding the equivalence between the deactivation rate of the ODE (3) and the dominant eigenvalue $\lambda_1(E)$ of the linear term $A(E)$.

Theorem 3.2 For any non-zero initial condition $x(0) \in \mathbb{R}_{>0}^n$, the Lyapunov characteristic exponent $\Lambda(E; x(0))$ of the ODE (3) is identical to the dominant eigenvalue $\lambda_1(E)$ of the linear term $A(E)$.

Proof From Theorem 3.1, it holds that $\lim_{t \rightarrow \infty} \|x(t)\| = 0$. This guarantees that for any sufficiently small neighborhood $U \subset \mathbb{R}^n$ of the origin, there exists a finite time $T \geq 0$ such that the solution satisfies $x(t) \in U$ for all $t \geq T$. Therefore, it is sufficient to evaluate the solution $x(t)$ within the domain U , because the solution $x(t)$ for $t \in [0, T)$ does not affect the value of $\Lambda(E; x(0))$ which is defined by the limit as $t \rightarrow \infty$ in (15). By shifting the time origin $t \leftarrow t - T$, we may assume without loss of generality that $x(0) \in U$ and proceed with the analysis for all $t \in [0, +\infty)$.

Within the domain U , we analyze the ODE (3) by reducing it to its normal form. Since the dominant eigenvalue $\lambda_1(E)$ is a simple root as shown in (7), there exists an invertible matrix $P \in \mathbb{C}^{n \times n}$ such that the coefficient matrix $A(E)$ is transformed into its Jordan normal form $\tilde{A}(E) := P^{-1}A(E)P$. Specifically, $\tilde{A}(E)$ gives a block diagonal structure separating the dominant eigenvalue $\lambda_1(E)$ from the others $\lambda_2(E), \dots, \lambda_n(E)$ [21, 22]:

$$\tilde{A}(E) = \begin{pmatrix} \lambda_1(E) & 0 \\ 0 & J(E) \end{pmatrix}, \tag{17}$$

where $J(E) \in \mathbb{C}^{(n-1) \times (n-1)}$ is the Jordan matrix associated with the remaining eigenvalues $\lambda_2(E), \dots, \lambda_n(E)$, and 0 denotes the zero matrix of appropriate dimensions (i.e., $1 \times (n-1)$ and $(n-1) \times 1$). Applying the linear transformation $x = Pu$ to the ODE (3), we have

$$\frac{d}{dt} u(t) = \tilde{A}(E)u(t) + \tilde{F}(u(t)) \tag{18}$$

where $\tilde{F}(u) := P^{-1}F(Pu)$ satisfies $\|\tilde{F}(u)\| = \mathcal{O}(\|u\|^2)$.

Next, we introduce a near-identity transformation $u = z + R(z)$ with $\|R(z)\| = \mathcal{O}(\|z\|^2)$. This transforms the ODE (18) into the Poincaré-Dulac normal form [29]:

$$\frac{d}{dt}z(t) = \tilde{A}(E)z(t) + N(z(t)). \quad (19)$$

Because of the block structure of $\tilde{A}(E)$ in (17), the linear part of the first component z_1 is decoupled from the other components $z_2, \dots, z_n$. The vector field $N(z)$ consists of resonant monomial terms [29]. That is, a term proportional to $\prod_{i=1}^n z_i^{m_i}$ exists in the k -th component $N_k(z)$ if and only if the following resonance condition is satisfied

$$\lambda_k(E) = \sum_{i=1}^n m_i \lambda_i(E), \quad k = 1, \dots, n \quad (20)$$

where $m_i \in \mathbb{Z}_{\geq 0}$ ($i = 1, \dots, n$) and $\sum_{i=1}^n m_i \geq 2$.

We analyze the resonance condition (20) for the component z_1 , corresponding to the dominant real eigenvalue $\lambda_1(E)$. From (7), $\lambda_1(E)$ is a negative real value satisfying $\lambda_1(E) \geq \operatorname{Re}(\lambda_i(E))$ for all $i = 1, \dots, n$. Suppose the resonance condition (20) holds for $k = 1$. Taking the real part of both sides of Eq. (20), we obtain

$$\lambda_1(E) = \sum_{i=1}^n m_i \operatorname{Re}(\lambda_i(E)) \leq \sum_{i=1}^n m_i \lambda_1(E) = \lambda_1(E) \sum_{i=1}^n m_i \quad (21)$$

Since $\sum_{i=1}^n m_i \geq 2$ and $\lambda_1(E) < 0$, it follows that

$$\lambda_1(E) \sum_{i=1}^n m_i \leq 2\lambda_1(E) \quad (22)$$

Combining (21) and (22), we obtain:

$$\lambda_1(E) \leq 2\lambda_1(E), \quad (23)$$

which implies that $0 \leq \lambda_1(E)$. This contradicts $\lambda_1(E) < 0$. Therefore, the resonance condition (20) for the component z_1 is not satisfied. Consequently, it holds that $N_1(z) \equiv 0$. From Eqns. (17) and (19), the component z_1 is given by

$$\frac{d}{dt}z_1(t) = \lambda_1(E)z_1(t), \quad \forall t \in [0, +\infty), \quad (24)$$

which implies that $z_1(t) = z_1(0)e^{\lambda_1(E)t}$ for all $t \in [0, +\infty)$. Note that $z_1(0) \neq 0$.

For the remaining components z_j ($j = 2, \dots, n$), let us define the vector $z_{\perp} := (z_2, \dots, z_n)^T$. The ODE for $z_{\perp}$ is described by:

$$\frac{d}{dt}z_{\perp}(t) = J(E)z_{\perp}(t) + N_{\perp}(z(t)) \quad (25)$$

By the variation of constants formula, the solution $z_{\perp}(t)$ to the ODE (25) is written as:

$$z_{\perp}(t) = e^{J(E)t}z_{\perp}(0) + \int_0^t e^{J(E)(t-\tau)}N_{\perp}(z(\tau))d\tau, \quad \forall t \in [0, +\infty), \quad (26)$$

where $N_{\perp}(z) := (N_2(z), \dots, N_n(z))^T$. The maximum real part of the eigenvalues in the Jordan block $J(E)$ is $\operatorname{Re}(\lambda_2(E))$. By properties of matrix exponentials [29], for any arbitrarily small $\epsilon > 0$, there exists a constant $C_{\epsilon} > 0$ such that

$$\|e^{J(E)t}\| \leq C_{\epsilon}e^{(\operatorname{Re}(\lambda_2(E)) + \epsilon)t}, \quad \forall t \in [0, +\infty). \quad (27)$$

Furthermore, from the inequality (14) and $\|N_{\perp}(z)\| = \mathcal{O}(\|z\|^2)$, it has been shown that

$$\|z(t)\| \leq C_0 e^{\lambda_1(E)t}, \quad \forall t \in [0, +\infty) \quad (28)$$

for some constant $C_0 > 0$. In addition, there exists a constant $K' > 0$ such that

$$\|N_\perp(z(t))\| \leq K' e^{2\lambda_1(E)t}, \quad \forall t \in [0, +\infty). \quad (29)$$

Taking the norm of (26) and applying these bounds (27)–(29), we obtain

$$\|z_\perp(t)\| \leq C_\epsilon e^{(\text{Re}(\lambda_2(E)) + \epsilon)t} \|z_\perp(0)\| + \int_0^t C_\epsilon e^{(\text{Re}(\lambda_2(E)) + \epsilon)(t-\tau)} K' e^{2\lambda_1(E)\tau} d\tau, \quad \forall t \in [0, +\infty). \quad (30)$$

Hence, we can show that there exists a constant $M > 0$ such that

$$\|z_\perp(t)\| \leq M e^{(\text{Re}(\lambda_2(E)) + \epsilon)t} + M e^{2\lambda_1(E)t}, \quad \forall t \in [0, +\infty). \quad (31)$$

Since $\text{Re}(\lambda_i(E)) \leq \lambda_1(E)$ ($i = 2, \dots, n$), for any $\epsilon > 0$, there exists a constant $M > 0$ such that:

$$|z_j(t)| \leq M e^{(\text{Re}(\lambda_2(E)) + \epsilon)t} + M e^{2\lambda_1(E)t}, \quad \forall t \in [0, +\infty). \quad (32)$$

Since $\text{Re}(\lambda_2(E)) < \lambda_1(E)$ due to Eq. (7), we can choose ϵ sufficiently small such that $\text{Re}(\lambda_2(E)) + \epsilon < \lambda_1(E)$. Using $|z_1(t)| = |z_1(0)| e^{\lambda_1(E)t}$ from (24), we can evaluate the ratio of their norms:

$$\begin{aligned} \frac{\|z_\perp(t)\|}{|z_1(t)|} &\leq \frac{M e^{(\text{Re}(\lambda_2(E)) + \epsilon)t} + M e^{2\lambda_1(E)t}}{|z_1(0)| e^{\lambda_1(E)t}} \\ &= \frac{M}{|z_1(0)|} \left(e^{(\text{Re}(\lambda_2(E)) + \epsilon - \lambda_1(E))t} + e^{\lambda_1(E)t} \right), \quad \forall t \in [0, +\infty). \end{aligned} \quad (33)$$

Since $\lambda_1(E) < 0$ and $\text{Re}(\lambda_2(E)) + \epsilon - \lambda_1(E) < 0$, it follows that:

$$\lim_{t \rightarrow \infty} \frac{\|z_\perp(t)\|}{|z_1(t)|} = 0 \quad (34)$$

Returning to the original variables $x(t)$ via $x(t) = Pu(t) = P(z(t) + R(z(t)))$, we consider the norm $\|x(t)\|$. First, for the variables $z(t) = (z_1(t), z_\perp^T(t))^T$, we have:

$$\|z(t)\| = |z_1(t)| \sqrt{1 + \left(\frac{\|z_\perp(t)\|}{|z_1(t)|} \right)^2}, \quad \forall t \in [0, +\infty). \quad (35)$$

From Eq. (34), the ratio $\|z_\perp(t)\|/|z_1(t)| \rightarrow 0$ as $t \rightarrow \infty$. This guarantees that for any small $\epsilon_1 > 0$, there exists $T_1 > 0$ such that:

$$|z_1(t)| \leq \|z(t)\| \leq (1 + \epsilon_1) |z_1(t)|, \quad \forall t > T_1. \quad (36)$$

Furthermore, since $\|R(z)\| = O(\|z\|^2)$ and $\lim_{t \rightarrow \infty} \|z(t)\| = 0$, the following limit holds:

$$\lim_{t \rightarrow \infty} \frac{\|z(t) + R(z(t))\|}{\|z(t)\|} = 1. \quad (37)$$

Therefore, for any small $\epsilon_2 \in (0, 1)$, there exists $T_2 > 0$ such that the norm of $u(t) = z(t) + R(z(t))$ is strictly bounded by:

$$(1 - \epsilon_2) \|z(t)\| \leq \|u(t)\| \leq (1 + \epsilon_2) \|z(t)\|, \quad \forall t > T_2. \quad (38)$$

Finally, because $x(t) = Pu(t)$ with an invertible matrix P , there exist positive constants c_m, C_M such that

$$c_m \|u(t)\| \leq \|x(t)\| \leq C_M \|u(t)\|. \quad (39)$$

By setting $T' = \max\{T_1, T_2\}$ and chaining these inequalities (36)–(39) with $|z_1(t)| = |z_1(0)| e^{\lambda_1(E)t}$, we see that for any arbitrarily small $\delta \in (0, 1)$, we can choose ϵ_1 and ϵ_2 such that:

$$c_m (1 - \delta) |z_1(0)| e^{\lambda_1(E)t} \leq \|x(t)\| \leq C_M (1 + \delta) |z_1(0)| e^{\lambda_1(E)t}, \quad \forall t > T'. \quad (40)$$

Substituting Eq. (40) into the Lyapunov characteristic exponent (15) and taking the logarithm, we obtain

$$\lim_{t \rightarrow \infty} \frac{\ln(c_m(1 - \delta)|z_1(0)|)}{t} + \lambda_1(E) \leq \Lambda(E; x(0)) \leq \lim_{t \rightarrow \infty} \frac{\ln(C_M(1 + \delta)|z_1(0)|)}{t} + \lambda_1(E) \quad (41)$$

Since it is clear that

$$\lim_{t \rightarrow \infty} \frac{\ln(c_m(1 - \delta)|z_1(0)|)}{t} = \lim_{t \rightarrow \infty} \frac{\ln(C_M(1 + \delta)|z_1(0)|)}{t} = 0, \quad (42)$$

we see from (41) that

$$\Lambda(E; x(0)) = \lambda_1(E). \quad (43)$$

This completes the proof. $\square$

4 Estimation of Deactivation Rates under Potentiostatic Conditions

Theorem 3.2 shows that the deactivation rate of the ODE (3) given by the Lyapunov characteristic exponent $\Lambda(E; x(0))$ is identical to the dominant eigenvalue $\lambda_1(E)$ of the linear term $A(E)$. Based on this equivalence, we define the deactivation regimes for the ODE (3) in the limit of $\Lambda(E; x(0))$ as the potential $E \rightarrow +\infty$ [36].

Definition 4.1 The qualitative behaviour of the deactivation rate $\Lambda(E; x(0)) (= \lambda_1(E))$ for the non-linear chemical reaction network undergoing dissolution as the potential $E \rightarrow +\infty$ can be decomposed into three types:

- (i) If $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$, the deactivation rate is said to be accelerated with respect to the potential E .
- (ii) If $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = 0$, the deactivation rate is said to be decelerated with respect to the potential E .
- (iii) If there exists a negative constant $\bar{\lambda}_1 \in (-\infty, 0)$ such that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = \bar{\lambda}_1$, the deactivation rate is said to be constant with respect to the potential E . $\square$

Before classifying these asymptotic behaviors, we characterize the algebraic structure of the coefficients of the characteristic polynomial of the matrix $A(E)$ given by:

$$f(\lambda) := \det(\lambda I_n - A(E)) = \sum_{i=0}^n a_i(E) \lambda^{n-i}, \quad \lambda \in \mathbb{C} \quad (44)$$

where $a_0(E) \equiv 1$, and from the Routh-Hurwitz Theorem [30], each coefficient $a_i(E)$ is a positive function with respect to $E > 0$ because the real parts of all roots of (44) are negative.

Let $G' = (\mathcal{V}', \mathcal{E}')$ be an augmented directed graph, where the vertex set $\mathcal{V}' = \{X_1, \dots, X_n\} \cup \{\emptyset\}$ incorporates the node representing a state external to the catalytic cycle, and the edge set $\mathcal{E}'$ contains all internal catalytic reactions in $\mathcal{E}$ as well as the dissolution pathways $X_k \rightarrow \emptyset$. In accordance with graph-theoretic definitions for Laplacian dynamics [21, 22], a subgraph T of G' is called a directed tree if it is connected and acyclic as an undirected graph. Furthermore, T is said to be rooted at a vertex $v \in \mathcal{V}'$ if v is the only vertex in T with no edge leaving it, which implies that every other vertex in T has exactly one outgoing edge, thereby forming a directed path from any vertex in T to the root v . An isolated single vertex with no edges trivially satisfies this condition and is considered a valid tree rooted at itself.

A directed forest F on G' is defined as a subgraph whose connected components are mutually disjoint directed trees. A spanning directed forest is one that contains all vertices of the graph G' . Let $\mathfrak{F}_i$ be the set of all spanning directed forests on G' that comprise exactly $n-i+1$ trees. This satisfies the topological condition that exactly one tree is rooted at the dissolution sink $\emptyset$, while the remaining $n-i$ trees are rooted

at distinct vertices chosen from $\{X_1, \dots, X_n\}$. Since the augmented graph G' contains $n + 1$ vertices and any forest $F \in \mathfrak{F}_i$ consists of $n - i + 1$ trees, every forest in $\mathfrak{F}_i$ contains exactly $(n + 1) - (n - i + 1) = i$ edges.

According to the All Minors Matrix Tree Theorem [25], each coefficient $a_i(E)$ enumerates the total weight of the spanning directed forests in $\mathfrak{F}_i$:

$$a_i(E) = \sum_{F \in \mathfrak{F}_i} W(F) \quad (45)$$

Let the weight of a directed forest F , denoted as $W(F)$, be the product of the weights of its constituent edges. Under the assumption of Butler-Volmer kinetics (6), the weight of each edge e is given by $k_e(E) = k_e^0 \exp(\alpha_e E)$. Therefore, the weight of the forest F , which consists of the product of exactly i edge weights, can be calculated as follows:

$$W(F) = \prod_{e \in F} k_e(E) = \prod_{e \in F} k_e^0 \exp(\alpha_e E) = \left(\prod_{e \in F} k_e^0 \right) \exp\left(\sum_{e \in F} \alpha_e E \right) \quad (46)$$

By defining the constant $C_F := \prod_{e \in F} k_e^0 > 0$ and the exponent sum $\gamma_F := \sum_{e \in F} \alpha_e$ (which corresponds to the sum of the electron transfer coefficients of the i edges in F), the weight of the forest is expressed as $W(F) = C_F \exp(\gamma_F E)$. Consequently, each coefficient $a_i(E)$ is expressed as a finite sum of exponential functions:

$$a_i(E) = \sum_{F \in \mathfrak{F}_i} C_F \exp(\gamma_F E) \quad (47)$$

By applying the graph-theoretic representation of the polynomial coefficients, we establish the necessary and sufficient conditions for the deactivation regimes of the ODE (3) based on the network topology.

Theorem 4.1 For the ODE (3) defined on a strongly connected graph G with at least one dissolution pathway, let $\mathfrak{F}_i$ be the set of spanning directed forests on the augmented graph G' comprising $n - i + 1$ trees. Let $\gamma_F := \sum_{e \in F} \alpha_e$ be the sum of the electron transfer coefficients of the edges in a forest F . Define the maximum exponent for the forest set $\mathfrak{F}_i$ as:

$$\gamma_{i,\max} := \max_{F \in \mathfrak{F}_i} \gamma_F, \quad i = 1, \dots, n. \quad (48)$$

Then, the asymptotic deactivation regimes of $\Lambda(E; x(0))$ as $E \rightarrow +\infty$ are classified as follows:

- (i) The deactivation rate is accelerated if and only if $\gamma_{n,\max} > \gamma_{n-1,\max}$.
- (ii) The deactivation rate is decelerated if and only if $\gamma_{n,\max} < \gamma_{n-1,\max}$.
- (iii) The deactivation rate is constant if and only if $\gamma_{n,\max} = \gamma_{n-1,\max}$.

Proof From Theorem 3.1 in [36], the asymptotic behavior of the dominant eigenvalue $\lambda_1(E)$ of the linear term $A(E)$ as $E \rightarrow +\infty$ is uniquely determined by the limit of the ratio $a_{n-1}(E)/a_n(E)$. Specifically, the following three conditions have been shown

- (i) $\lim_{E \rightarrow +\infty} \lambda_1(E) = -\infty$ if the limit holds that:

$$\lim_{E \rightarrow +\infty} \frac{a_{n-1}(E)}{a_n(E)} = 0. \quad (49)$$

- (ii) $\lim_{E \rightarrow +\infty} \lambda_1(E) = 0$ if the limit holds that:

$$\lim_{E \rightarrow +\infty} \frac{a_{n-1}(E)}{a_n(E)} = +\infty. \quad (50)$$

(iii) There exists a negative constant $\bar{\lambda}_1 \in (-\infty, 0)$ such that $\lim_{E \rightarrow +\infty} \lambda_1(E) = \bar{\lambda}_1$ if the finite limit exists:

$$\lim_{E \rightarrow +\infty} \frac{a_{n-1}(E)}{a_n(E)} \in (0, +\infty). \quad (51)$$

By substituting the graph-theoretic representation (47), the ratio is given by:

$$\frac{a_{n-1}(E)}{a_n(E)} = \frac{\sum_{F \in \mathfrak{F}_{n-1}} C_F \exp(\gamma_F E)}{\sum_{F \in \mathfrak{F}_n} C_F \exp(\gamma_F E)}. \quad (52)$$

Because all coefficients C_F are positive constants, the asymptotic behavior of this ratio as $E \rightarrow +\infty$ is entirely dominated by the terms with the maximum exponents in the numerator $\gamma_{n-1, \max}$ and the denominator $\gamma_{n, \max}$. Thus, by evaluating the limit (52) and using $\Lambda(E; x(0)) = \lambda_1(E)$, we obtain the corresponding three regimes:

- (i) If $\gamma_{n, \max} > \gamma_{n-1, \max}$, we have Eq. (49), which implies that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$.
- (ii) If $\gamma_{n, \max} < \gamma_{n-1, \max}$, we have Eq. (50). This means that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = 0$.
- (iii) If $\gamma_{n, \max} = \gamma_{n-1, \max}$, we obtain Eq. (51), which implies that there exists a negative constant $\bar{\lambda}_1 \in (-\infty, 0)$ such that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = \bar{\lambda}_1$.

This completes the proof. $\square$

Based on the graph-theoretic representation, we first derive an upper bound for the difference between the maximum exponents $\gamma_{n, \max}$ and $\gamma_{n-1, \max}$. This result provides a sufficient condition for the deactivation rate to not accelerate.

Corollary 4.1 For the ODE (3) defined on a strongly connected graph G with at least one dissolution pathway, let $F_n^* \in \mathfrak{F}_n$ be the tree forest corresponding to the maximum exponent $\gamma_{n, \max}$. The following inequality holds :

$$\gamma_{n, \max} - \gamma_{n-1, \max} \leq \min_{e \in F_n^*} \alpha_e. \quad (53)$$

Moreover, if F_n^* contains at least one reaction step e such that $\alpha_e = 0$, then $\gamma_{n, \max} \leq \gamma_{n-1, \max}$. This means that the deactivation rate is not accelerated, that is, $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) \neq -\infty$.

Proof Let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest. By the definition of a directed tree, every node in F_n^* except the root $\emptyset$ has an out-degree of exactly 1. We select an arbitrary edge $e \in F_n^*$. By deleting this single edge e from F_n^* , the subgraph $F_{n-1} := F_n^* \setminus \{e\}$ is separated into two disjoint directed trees: one rooted at $\emptyset$, and the other rooted at the tail node of the deleted edge e . Therefore, by definition, F_{n-1} is a spanning directed forest comprising 2 trees, which guarantees $F_{n-1} \in \mathfrak{F}_{n-1}$. The exponent sum of this forest F_{n-1} is given by

$$\gamma_{F_{n-1}} = \gamma_{F_n^*} - \alpha_e = \gamma_{n, \max} - \alpha_e. \quad (54)$$

Because $\gamma_{n-1, \max}$ is defined as the maximum exponent over all forests in $\mathfrak{F}_{n-1}$, we see that

$$\gamma_{n-1, \max} \geq \gamma_{F_{n-1}} = \gamma_{n, \max} - \alpha_e. \quad (55)$$

Rearranging the inequality, we obtain

$$\gamma_{n, \max} - \gamma_{n-1, \max} \leq \alpha_e. \quad (56)$$

Since this inequality holds for any arbitrarily chosen edge $e \in F_n^*$, it holds for the edge with the minimum electron transfer coefficient α_e . Thus, we obtain the bound (53). Furthermore, if there exists at least one reaction step in F_n^* such that its electron transfer coefficient is $\alpha_{\bar{e}} = 0$, by substituting $\min_{e \in F_n^*} \alpha_e \leq \alpha_{\bar{e}} = 0$ into (53), we have $\gamma_{n, \max} - \gamma_{n-1, \max} \leq 0$.

This completes the proof. $\square$

Applying this result to the dissolution pathways, we show that the deactivation rate is not accelerated if all dissolution steps are potential-independent.

Corollary 4.2 For the ODE (3) defined on a strongly connected graph G with at least one dissolution pathway, suppose that $\alpha_d = 0$ for all dissolution edges d . Then, $\gamma_{n,\max} \leq \gamma_{n-1,\max}$. That is, the deactivation rate is not accelerated, that is, $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) \neq -\infty$.

Proof Let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest corresponding to the maximum exponent $\gamma_{n,\max}$. By definition, F_n^* consists of exactly one tree rooted at the dissolution sink $\emptyset$. This implies that F_n^* must contain at least one dissolution edge. We select an arbitrary dissolution edge e_d in F_n^* . By deleting this single edge e_d from F_n^* , the subgraph $F_{n-1} := F_n^* \setminus \{e_d\}$ is separated into two disjoint directed trees: one rooted at $\emptyset$, and the other rooted at the tail node of the deleted edge e_d . Therefore, by definition, F_{n-1} is a spanning directed forest comprising 2 trees, which guarantees $F_{n-1} \in \mathfrak{F}_{n-1}$. The exponent sum of this forest F_{n-1} is given by:

$$\gamma_{F_{n-1}} = \gamma_{F_n^*} - \alpha_{e_d} = \gamma_{n,\max} - \alpha_{e_d}. \quad (57)$$

Since, by the assumption, we have $\alpha_{e_d} = 0$, we obtain $\gamma_{F_{n-1}} = \gamma_{n,\max}$. Because $\gamma_{n-1,\max}$ is defined as the maximum exponent over all forests in $\mathfrak{F}_{n-1}$, it follows that

$$\gamma_{n-1,\max} \geq \gamma_{F_{n-1}} = \gamma_{n,\max}, \quad (58)$$

which implies that

$$\gamma_{n,\max} - \gamma_{n-1,\max} \leq 0. \quad (59)$$

By Theorem 4.1, this inequality implies that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) \neq -\infty$.

This completes the proof. $\square$

Conversely, we provide a sufficient condition for the acceleration of the deactivation rate, which occurs when all internal catalytic steps share the same uniform potential sensitivity.

Corollary 4.3 For the ODE (3) defined on a strongly connected graph G with at least one dissolution pathway, suppose that $\alpha_e = \alpha_{int} > 0$ for all $e \in \mathcal{E}$, and that $\alpha_d > 0$ for all dissolution pathways d . Then, the deactivation rate is accelerated, that is, $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$.

Proof Let $F_{n-1}^* \in \mathfrak{F}_{n-1}$ be a spanning directed forest corresponding to the maximum exponent $\gamma_{n-1,\max}$. By definition, F_{n-1}^* consists of two disjoint directed trees: one rooted at the dissolution sink $\emptyset$, denoted by $T_\emptyset$, and the other rooted at an intermediate species X_k , denoted by T_k . Because the graph G is strongly connected, there exists a directed path from X_k to $\emptyset$. Let $e_{add} = X_k \rightarrow X_j$ be the first edge on the shortest such path.

If $X_j \in T_\emptyset$, the addition of e_{add} to F_{n-1}^* directly connects T_k to $T_\emptyset$ without creating any cycles. The graph is a spanning directed forest $F_n \in \mathfrak{F}_n$ comprising one tree. Its exponent sum is $\gamma_{F_n} = \gamma_{n-1,\max} + \alpha_{e_{add}}$. Since $\alpha_{e_{add}} > 0$ by assumption, we obtain $\gamma_{n,\max} \geq \gamma_{F_n} > \gamma_{n-1,\max}$.

If $X_j \in T_k$, the addition of e_{add} creates a unique cycle within T_k . In order to restore a 2-tree forest, we remove the original outgoing edge e_{remove} of node X_j in T_k . Since both e_{add} and e_{remove} are edges in $\mathcal{E}$, they share the same electron transfer coefficient: $\alpha_{e_{add}} = \alpha_{e_{remove}} = \alpha_{int}$. Thus, the new graph is a valid forest $F'_{n-1} \in \mathfrak{F}_{n-1}$ rooted at the tail of e_{remove} , and its exponent sum remains unchanged:

$$\gamma_{F'_{n-1}} = \gamma_{n-1,\max} + \alpha_{e_{add}} - \alpha_{e_{remove}} = \gamma_{n-1,\max}. \quad (60)$$

By recursively applying this edge-shifting operation, the root of the second tree moves along the shortest path toward $T_\emptyset$ while maintaining the maximal weight $\gamma_{n-1,\max}$. Because the path length is finite, this process must terminate in the first case, where an edge e^* connecting to $T_\emptyset$ is added. This guarantees the existence of a forest $F_n \in \mathfrak{F}_n$ with $\gamma_{F_n} = \gamma_{n-1,\max} + \alpha_{e^*}$. Since $\alpha_{e^*} \geq \min(\alpha_{int}, \alpha_d) > 0$, we conclude that:

$$\gamma_{n,\max} \geq \gamma_{F_n} > \gamma_{n-1,\max}. \quad (61)$$

By Theorem 4.1, this inequality implies that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$.

This completes the proof. $\square$

We provide a sufficient condition for the deceleration of the deactivation rate in networks possessing a single dissolution pathway.

Corollary 4.4 For the ODE (3) defined on a strongly connected graph G with exactly one dissolution pathway e_d from an intermediate species X_k , let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest corresponding to the maximum exponent $\gamma_{n,\max}$, and the subgraph $T_k^* := F_n^* \setminus \{e_d\}$ be a directed tree of the species rooted at X_k . Suppose there exists an outgoing edge $e_{out} \in \mathcal{E}$ from X_k such that its addition to T_k^* creates a cycle C , and there exists an edge $e_{remove} \in C$ satisfying

$$\alpha_d - \alpha_{e_{out}} + \alpha_{e_{remove}} < 0. \quad (62)$$

Then, the deactivation rate is decelerated, that is, $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = 0$.

Proof Let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest corresponding to the maximum exponent $\gamma_{n,\max}$. Because e_d is the edge connecting the species to the dissolution sink $\emptyset$, F_n^* must contain e_d . Thus, its exponent sum is given by $\gamma_{n,\max} = \gamma_{T_k^*} + \alpha_d$, where $\gamma_{T_k^*}$ is the exponent sum of the tree T_k^* rooted at X_k .

By adding the edge e_{out} to T_k^* , a unique cycle C is created. In order to break this cycle and form a new tree, we remove the edge $e_{remove} \in C$. This operation gives a new spanning tree T_{new} rooted at the tail node of e_{remove} . By pairing T_{new} with the isolated root node $\emptyset$, we construct a spanning directed forest $F_{n-1} \in \mathfrak{F}_{n-1}$ comprising 2 trees. The exponent sum of this forest F_{n-1} is given by

$$\gamma_{F_{n-1}} = \gamma_{T_k^*} + \alpha_{e_{out}} - \alpha_{e_{remove}}. \quad (63)$$

Because $\gamma_{n-1,\max}$ is defined as the maximum exponent over all forests in $\mathfrak{F}_{n-1}$, we see that

$$\gamma_{n-1,\max} \geq \gamma_{F_{n-1}} = \gamma_{T_k^*} + \alpha_{e_{out}} - \alpha_{e_{remove}}. \quad (64)$$

By the assumption $\alpha_d - \alpha_{e_{out}} + \alpha_{e_{remove}} < 0$, we have $\alpha_d < \alpha_{e_{out}} - \alpha_{e_{remove}}$. Adding $\gamma_{T_k^*}$ to both sides of this inequality, it follows that

$$\gamma_{n,\max} = \gamma_{T_k^*} + \alpha_d < \gamma_{T_k^*} + \alpha_{e_{out}} - \alpha_{e_{remove}} \leq \gamma_{n-1,\max}. \quad (65)$$

Thus, we obtain the inequality $\gamma_{n,\max} < \gamma_{n-1,\max}$. By Theorem 4.1, the inequality implies that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = 0$.

This completes the proof. $\square$

We establish a condition for accelerated deactivation in networks with a single dissolution pathway

Corollary 4.5 For the ODE (3) defined on a strongly connected graph G with exactly one dissolution pathway e_d originating from an intermediate species X_k , suppose that $\alpha_e > 0$ for all $e \in \mathcal{E}$. Let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest corresponding to the maximum exponent $\gamma_{n,\max}$. Here, its exponent sum is given by $\gamma_{n,\max} = \gamma_{T_k^*} + \alpha_d$, where $\gamma_{T_k^*}$ is the exponent sum of the tree $T_k^* := F_n^* \setminus \{e_d\}$ on $\mathcal{E}$. Furthermore, let $\gamma_{\mathcal{E},\max}$ be the maximum exponent sum among all 2-tree forests $F \in \mathfrak{F}_{n-1}$ that do not contain e_d (i.e., $F \subset \mathcal{E}$), and let $\gamma_{\mathcal{E},2}^*$ be the maximum exponent sum on $\mathcal{E}$ among all 2-tree forests $F \in \mathfrak{F}_{n-1}$ that contain e_d . Suppose that $\gamma_{T_k^*}^* > \gamma_{\mathcal{E},2}^*$. If the dissolution potential sensitivity α_d satisfies

$$\alpha_d > \gamma_{\mathcal{E},\max} - \gamma_{T_k^*}^*, \quad (66)$$

then the deactivation rate is accelerated, that is, $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$.

Proof Let $F_n^* \in \mathfrak{F}_n$ be the spanning directed forest corresponding to the maximum exponent $\gamma_{n,\max}$. Because e_d is the only edge connecting the species to the sink $\emptyset$, F_n^* contains e_d . Thus, its exponent sum is $\gamma_{n,\max} = \gamma_{T_k^*} + \alpha_d$.

We evaluate the maximum exponent $\gamma_{n-1,\max}$ for forests $F_{n-1} \in \mathfrak{F}_{n-1}$ comprising 2 trees. We classify F_{n-1} into two cases based on whether it contains the dissolution edge e_d .

If F_{n-1} does not contain e_d , the tree rooted at $\emptyset$ is the isolated node $\emptyset$, and the remaining edges form a spanning directed tree on $\mathcal{E}$. By definition, the maximum exponent sum for this first case is $\gamma_{\mathcal{E},\max}$.

If F_{n-1} contains e_d , its subgraph on $\mathcal{E}$ forms a 2-tree forest with one tree rooted at X_k . By definition, the maximum exponent sum on $\mathcal{E}$ for this second case is $\gamma_{\mathcal{E},2}^*$. Thus, the total exponent sum of this second case is bounded by $\gamma_{\mathcal{E},2}^* + \alpha_d$. Since $\gamma_{T_k^*} > \gamma_{\mathcal{E},2}^*$ by assumption, this total exponent sum is less than $\gamma_{T_k^*} + \alpha_d = \gamma_{n,\max}$.

Therefore, the condition $\gamma_{n-1,\max} \geq \gamma_{n,\max}$ can occur through the first case. To guarantee the inequality $\gamma_{n,\max} > \gamma_{n-1,\max}$, it is sufficient to require that

$$\gamma_{T_k^*} + \alpha_d > \gamma_{\mathcal{E},\max}, \quad (67)$$

which implies that $\alpha_d > \gamma_{\mathcal{E},\max} - \gamma_{T_k^*}$. By Theorem 4.1, it follows that $\lim_{E \rightarrow +\infty} \Lambda(E; x(0)) = -\infty$.

This completes the proof. $\square$

5 Applications

6 Conclusion

Acknowledgments

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